BASANT BHATTARAI

SENIOR DATA ENGINEER / AI ENGINEER / DATA & AI ARCHITECT

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PROFILE

Senior data engineer and data and AI architect with 9+ years in industry. At UXCam, owns the architecture of a data platform processing 50 TB and 100M+ events a day, leads a team of 5 engineers, moved it from OLTP to a big data lakehouse, and designs the agentic AI platform built into it: 40% lower infrastructure cost, 50% faster processing, 60% faster analytics delivery, 99.9% uptime. Sets the target architecture, chooses platforms on evidence and designs the data model with its governance and SOC 2 controls. Open to Senior or Lead Data Engineer, AI Engineer and Data & AI Architect roles.

EXPERIENCE

Senior Data Engineer | UXCam

Feb 2024 - present

- Own the architecture, reliability, data quality and governance of a platform processing 50 TB and 100M+ events a day on Kafka, Spark, Iceberg, Hive, Trino, dbt and Airflow: freshness objectives, alerting, 99.9% uptime, runbooks, data contracts, a named owner per table, and the SOC 2 criteria met for the platform and its infrastructure through every audit.
- Lead a team of 5 engineers on the data platform, mentoring through pairing, code review and design feedback; delivery time down 30%.
- Design and build the agentic AI platform inside the data platform, with security, compliance and access control to data and internal systems designed in: RAG pipelines, MCP servers, LangGraph and Google ADK agents with guardrails and evaluation, output landing as typed, provenanced rows: analytics delivery 60% faster, 75% less analyst effort.
- Designed and built an LLM video-analysis platform that describes each session recording for in-depth product intelligence.
- Run the architecture and schema review every new store, pipeline and agent workflow passes before production, and operate the SQL and NoSQL databases underneath: replication, recovery, access, tuning, upgrades.

Data Engineer | UXCam

Feb 2020 - Feb 2024

- Led the move from OLTP to a big data lakehouse for lower cost, faster queries, scale and deeper analytics: tables in Iceberg, Hive and other formats with partition and schema-evolution rules, and a producer and consumer that queue Spark compaction of small files.
- Rebuilt the core ETL layer in Spark and PySpark with Airflow orchestration, cutting processing time for 50 TB/day by 50% and unblocking real-time product dashboards.
- Made backfills and replays safe to re-run: idempotent loads, explicit watermarks for late data, and schema evolution designed in.
- Cut p95 latency 50% on Trino, ClickHouse, Citus and TimescaleDB serving 100M+ queries/day, through performance tuning and query optimisation: partitioning, clustering and model redesign.
- Led the migration of the data and database infrastructure to Kubernetes on EKS, Spark clusters included: 40% lower infrastructure cost and autoscaling that held under peak load.

Project Leader and Research Developer | SVCET, JNTU Anantapur

Nov 2018 - Mar 2019

- Led a team of 4 engineers to build an intelligent dialogue system with deep NLP: a sequence-to-sequence model of two RNNs in TensorFlow and Python, trained on movie-dialogue and Twitter corpora, with a Flask web interface; delivered 2 weeks early.

Software Developer Intern | SV Technology, India

2017 - 2020

- Built Python backend services in Flask and Django serving 10K+ users, at 85% test coverage.
- Designed PostgreSQL schemas, stored procedures and reports, set up Jenkins CI, and began the move into data engineering.

SELECTED PROJECTS

- **Database scale-out and migration:** a Postgres application database and a pgvector store moved to Citus as each outgrew one server; an RDS database moved to a self-managed Citus cluster; a ClickHouse cluster designed to its workload.
- **Reliability and cost:** PostgreSQL and TimescaleDB clusters with replication, pgBackRest and PgPool-II; a database recovered from its raw files; an RDS and Aurora fleet tuned and failover-tested; AWS database spend analysed for savings.
- **Data platforms and AI:** an analytics lakehouse on S3, Trino, Airflow and Glue, delivered running with access control, HA and DR; frameworks for LLM applications and agentic workflows, architecture to tested Python; a failing classifier rebuilt on Databricks.

SELECTED OPEN SOURCE

- **data-platform-architecture** and **ai-platform-architecture** (reference architectures): a SaaS company leaving Postgres, measured: 18 decision records, a Kimball model, SOC 2 readiness, a cost model and a Databricks proof of concept; then its AI usage assistant.
- **spark-sql-kinesis** (maintainer): Kinesis source and sink for Spark 4.2 on AWS SDK v2, on Maven Central; an idle shard costs one GetRecords call, not a multi-second floor per batch.
- **governed-agent** and **evalproof** (LangGraph, MCP, LLM evaluation): answers matched a SQL-computed key on 48/54 questions, and 0 of 5 planted prompt injections led to a proposed action; evalproof fails CI only on LLM-judge drops both large and significant.
- Also **distributed-db-lab** (Citus sharding and online rebalance, against TimescaleDB and ClickHouse), data-stack-lab, streaming, real-time ML, retrieval, data contracts and GitOps delivery. github.com/technobasant

SKILLS

Data and AI architecture

Target architectures and platform selection, conceptual to physical modelling, data warehouse and lakehouse modelling (Kimball, Type 2 history), ClickHouse and Citus layouts, contracts in CI, decision records and architecture documentation, SOC 2 and GDPR erasure by design, tenant isolation outside the model, semantic layers, release gates, cost models.

Data platforms

Python, SQL and Scala; Kafka and Kinesis ingestion; Spark/PySpark, including the Spark 4 migration; Airflow; Iceberg, Hive Metastore, dbt and lakehouse modelling; Databricks with Unity Catalog; Snowflake (working knowledge); CI/CD with GitHub Actions, Kubernetes, Docker and Terraform; data quality, lineage and recovery; Prometheus and Grafana.

AI engineering

LangGraph, LangChain, Google ADK, MCP servers, Pydantic, Milvus and MLflow; prompt and structured-output design, guardrails, access control to data and internal systems, retrieval, evaluation on Qwen3, Llama 3.3, gpt-oss and Databricks Genie, provenance, human review and fallbacks; deep-learning NLP in TensorFlow since 2018.

Database lifecycle

PostgreSQL, ClickHouse, Trino, Citus, TimescaleDB, Redis and MongoDB; partitioning, replication, restore and high availability.

Cloud

AWS (production): S3, EKS (IRSA), Lambda, Glue, Athena, Kinesis, EMR, RDS | **GCP** (working knowledge): BigQuery, Dataflow, Dataproc, Datastream, Pub/Sub | **Azure** (working knowledge): Synapse, Data Factory, Databricks, ADLS, Event Hubs

EDUCATION AND LANGUAGES

Bachelor of Technology in Computer Science and Engineering

JNTU Anantapur (SVCET, an autonomous affiliated college), India | 2015 - 2019

English: C1 | Nepali: native

Selected work, writing and case studies: basantbhattarai.com.np/work/